

Prompt playground: versions, the 10 MB path, sharing

front: what to ask, what to type · Google Doc, no diagram · 5–15 min per section

Anthropic questions this is: SD Q2 prompt playground (1point3acres, 11 discussions, "the round that surprised candidates"): Google Doc with client ↔ web server ↔ DB pre-drawn, do not draw architecture; anchors 1M users, 10k DAU, 80/20, under 1 RPS (do not shard), 1 KB average, **10 MB max**; drill = schema in 90 s → flows → sharing scopes + revocation vs signed URLs → viral spike → the 10 MB path → many tabs → export to a real API call · **prachub** "prompt playground", "prompt sharing product", "AI playground for very large prompts" · **hellointerview** platform flavour (streaming, side-by-side, batch runs) = your deferred list · Kyle's seven questions = the same probes. **Extra Anthropic probes**: say the traffic trap early, revocation vs signed-URL expiry, many tabs (branch or merge).

1 REQUIREMENTS 0–5 min · questions, then six sentences

- **Ask**: browser product or API? single user or teams? one model? what is a prompt: one message, or system + user + few-shot? how big can a prompt be? keep outputs? sharing in scope, to whom?
- **A user can**: create a prompt · edit it · **run it and see the output** · save versions, history, diffs · rename / organise (threads) · share by link · delete.
- **Deferred, one sentence**: model + params picker, variables/templates, tags, ratings, datasets/evals, batch runs, team roles. Name them, leave them.
- **Technical challenges**: traffic tiny (1M users · 10k DAU · <1 RPS · 80/20) → **no sharding** · prompts 1 KB avg, 10 MB max ≈ 2.5M tokens → the blob path is the design · immutable versions make history, diffs, sharing, caching cheap · sharing needs scopes, revocation, viral reads.

2 CORE ENTITIES 5–9 min · three core, with fields

- **Prompt**(id, owner_id, name, latest_version_id, updated_at)
- **PromptVersion**(id, prompt_id, parent_version_id, body | body_ref, hash, size) — immutable; inline under 64 KB, object storage above
- **Output**(id, version_id, model, content_ref, tokens_in/out, latency_ms)
- **ShareLink**(token, prompt_id, version_id?, scope public|link|account, role viewer|editor, expires_at, revoked_at) · **AclEntry**(prompt_id, principal_id, role) when teams are in scope · op_log append-only
- Indexes: (owner_id, updated_at desc) · (prompt_id, created_at desc) · unique(token)
- Not entities: run (= Output row), thread (a name), model (a field), member/dataset/eval (not agreed in §1).

3 API 9–13 min · one endpoint per need

- POST /prompts {name} · PATCH /prompts/{id} {name} · DELETE /prompts/{id}
- POST /prompts/{id}/versions {parent_version_id, body | body_ref} · GET /prompts/{id}/versions · GET /versions/{a}/diff/{b}
- POST /versions/{id}/run → SSE stream, then Output stored
- POST /uploads {size, sha256} → **presigned multipart URL**; client uploads to object storage, then creates the version with body_ref
- POST /prompts/{id}/share {scope, role, expires_in} · DELETE /share/{token} · GET /s/{token} → rendered view

4 DATA FLOW 13–20 min · small, large, shared

- **Small**: autosave draft locally → Save = immutable version (inline) → Run sends version id → model gateway streams tokens → Output stored → history shows versions + diffs.
- **Over 10 MB**: client gates + shows token count → presigned multipart upload straight to object storage (resumable, per-part hash) → version row holds body_ref + hash → service never buffers the body → reads use range-GET into a virtualised editor. Running it exceeds any context window: say so, reject clearly or offer chunk-and-summarise.
- **Share**: owner creates link (scope, role) → reader hits /s/{token} → check revoked_at + expiry → serve the version; public + immutable → CDN-cacheable.

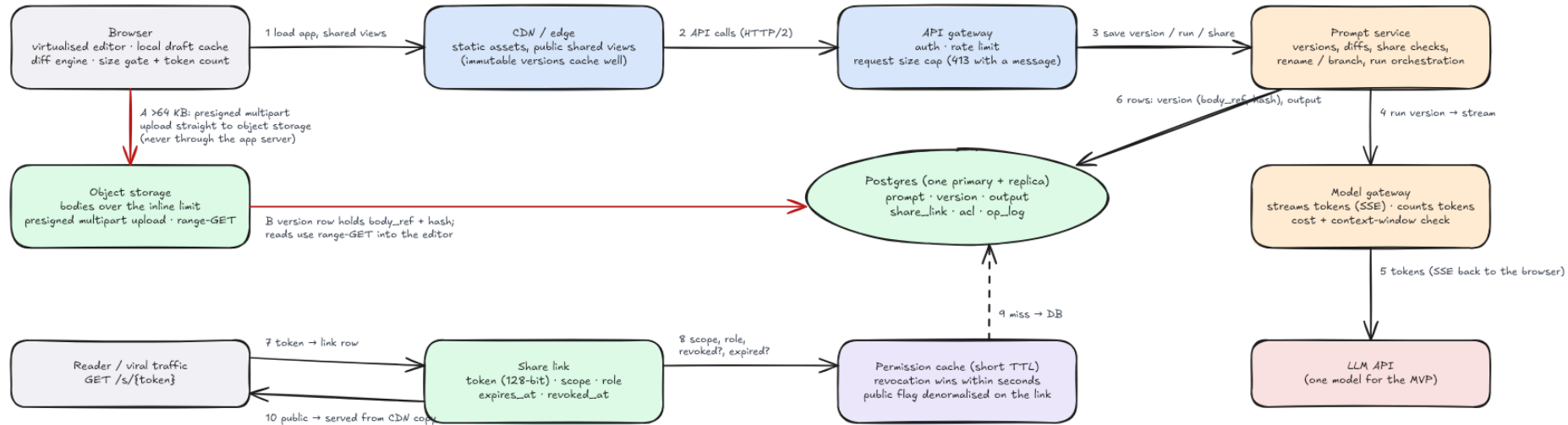
5 HIGH-LEVEL DESIGN 20–33 min · in words; draw only if invited

- Browser (virtualised editor, local cache, diff engine) → CDN/edge → API gateway (auth, **size cap**, rate limit) → prompt service (versions, diffs, share checks, runs) → Postgres (prompt, version, output, share_link, acl) · object storage (bodies over the limit, presigned upload, range-GET) · model gateway (SSE, token count, cost).
- One primary + read replica; **no sharding** at under 1 RPS. Say it in minute one.
- **10 MB hop by hop**: client timeouts/memory · gateway 413 · app server buffers → memory pressure · Postgres row/TOAST limits, bloat · model context window + cost. **Fix**: cap at the edge with a message · presigned upload · body_ref · compression (3–5×) · chunked resumable · range-GET.

6 DEEP DIVES 33–48 min · Kyle's order, then Anthropic's extras

- **Renames / divergence**: user rename, auto-name from first line or summary; versions have parents → edit of an old version = **branch**; branch-or-merge dialog, never overwrite.
- **Uploads without overloading**: presigned URL, bytes go client → object storage; service sees metadata + completion; multipart, per-part hash, dedupe by hash.
- **Frontend**: virtualised editor · lazy-load history/outputs · IndexedDB cache + prefetch · autosave · size gate + token counter · CDN for shared views · HTTP/2 (gRPC service-to-service only).
- **Upload size**: diffs vs parent + **checkpoint** every N versions · compress · chunk · dedupe.
- **Sharing**: audience first (private, team ACL, link-only, public) · 128-bit token, unique index, expiry · revocation = revoked_at + short-TTL permission cache; blob URLs signed 5–15 min (two-layer trade-off) · viral: immutable → CDN copy, public flag denormalised on the link row.
- **Many tabs**: shared in-browser cache, optimistic writes, conflict → branch or merge. **Bow**: tiny traffic, the design is the blob path + sharing semantics; every deferred feature is one sentence away.

The board, if the interviewer invites a diagram back: the answer, as it should look on the shared screen



Failure / probes: 10 MB prompt through the app server (timeouts, 413, memory, Postgres TOAST, context window) → cap at the edge, presigned upload, body_ref, compression, chunked resumable transfer, range-GET. Diverging edits → versions have parents, so an edit of an old version is a branch; show branch-or-merge, never overwrite. Revoked link → revoked_at checked through a short-TTL cache; blob URLs signed for 5–15 min.

Numbers to say: 1M users · 10k DAU · under 1 RPS (do not shard) · 1 KB average, 10 MB max ≈ 2.5M tokens (beyond any context window) · ~10 prompts per user ≈ 10 GB hot · inline body under 64 KB, object storage above · text compresses 3–5× · checkpoint every 20 diffs.

Frontend checklist: virtualised editor · lazy-load history and outputs · IndexedDB cache + prefetch next version · autosave drafts · client-side size gate + token counter · diffs, compression, chunking, dedupe on upload · CDN for shared views · HTTP/2 (gRPC only service-to-service).

Legend

 client / reader
 edge, gateway
 service
 store / link
 cache
 model

solid = requests red = the 10 MB path dashed = cache miss